RUIXIANG (RYAN) TANG

CONTACT INFORMATION

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RESEARCH INTERESTS

My research focuses on developing algorithms and systems that enhance trustworthiness throughout the AI lifecycle. I pursue this goal by formalizing key trustworthiness objectives, including robustness, interpretability, and safety, and by developing principled algorithms with measurable guarantees and transparent failure modes. I also collaborate closely with biologists and health informaticians to translate trustworthy learning methods to high-impact problems in biology and medicine.

ACADEMIC EMPLOYMENTS

Rutgers University–New Brunswick, New Jersey, USA *September 2024 - Present*
Assistant Professor
Department of Computer Science

EDUCATION

Rice University, Houston, USA *August 2021 - August 2024*
Ph.D. in Computer Science
Department of Computer Science Advisor: Dr. Xia (Ben) Hu

Texas A&M University, College Station, USA *August 2019 - August 2021*
Ph.D. Student in Computer Science (Transfer Out)
Department of Computer Science and Engineering Advisor: Dr. Xia (Ben) Hu

Tsinghua University, Beijing, China *September 2014 - June 2019*
Bachelor of Science in Automation
Department of Automation Advisor: Dr. Jiwen Lu

SELECTED HONORS, AWARDS, & FELLOWSHIPS

- Amazon Research Award, Spring 2026
- 2026 IEEE International Conference on Multimedia and Expo Best Paper Candidate
- AMD University Program: AI & HPC Cluster Allocation Award 2026
- OpenAI Researcher Access Program 2025
- Hershel M. Rich Invention Award 2024
- AMIA 2023 Best Student Paper Award
- CIKM 2023 Best Demo Paper Honorable Mention Award
- AMIA 2022 Best Student Paper (Shortlist) Award
- Microsoft Research Asia Rising Star Intern Award
- Outstanding Freshman Scholarship, Tsinghua University (Top 10%)

- The Silver Medal of IGEM (International Genetically Engineered Machine)
- China High School Biology Olympiad (Ranked Seventh Nationally)

PUBLICATIONS

60+ peer-reviewed papers published in top AI conferences and journals
ICME Best Paper Candidate
AMIA Best Student Paper
CIKM Best Demo Paper Honorable Mention
Communications of the ACM Cover Article
5,000+ citations, h-index 29 (Google Scholar).
* denotes an equal contribution, † denotes the corresponding author.

[NeurIPS' 26] Boxuan Zhang*, Jianing Zhu*, Zeru Shi, Dongfang Liu, **Ruixiang Tang**†. "AgentForesight: Online Auditing for Early Failure Prediction in Multi-Agent Systems." In *Conference on Neural Information Processing Systems, 2026.*

[NeurIPS' 26] Runjia Zeng, Hang Hua, Yiyang Liu, Zhiqiang Tao, **Ruixiang Tang**, Qifan Wang, Cheng Han, Dongfang Liu. "On the Token Value Inequality in Efficient Reasoning." In *Conference on Neural Information Processing Systems, 2026.*

[EMNLP' 26] Jiaqian Li, Yanshu Li, Boxuan Zhang, **Ruixiang Tang**, Kuan-Hao Huang. "TRACES: Proactive Safety Auditing for Multi-Turn LLM Agents via Trajectory-State Modeling." In *Empirical Methods in Natural Language Processing, EMNLP Main, 2026.*

[COLM' 26] Yihao Quan, Zeru Shi, Jinman Zhao, **Ruixiang Tang**†. "From Plausible to Grounded: Reinforcing Structured Consistency in Video MLLMs." In *Conference on Language Modeling, 2026.*

[COLM' 26] Rui Wu, **Ruixiang Tang**†. "From Rebound to Remedy: Understanding and Mitigating Reward Hacking via Representation Engineering." In *Conference on Language Modeling, 2026.*

[ECCV' 26] Yanshu Li, Jiaqian Li, Kuai Yu, Xi Xiao, Dongfang Liu, Tianyang Wang, **Ruixiang Tang**†. "Personalize Your Large Vision-language Models With In-context Prompt Tuning." In *European Conference on Computer Vision, 2026.*

[ECCV' 26] Feng He, Zhenting Wang, Qifan Wang, Qiang Guan, Dongfang Liu, **Ruixiang Tang**, Qiankun Li. "HIVE: Understanding Post-hallucination Reasoning in Vision Language Models." In *European Conference on Computer Vision, 2026.*

[ECCV' 26] Liwei Che*, Zhiyu Xue*, Yihao Quan*, Benlin Liu, Zeru Shi, **Ruixiang Tang**†, Ranjay Krishna, Vladimir Pavlovic†. "Learning from Primitive: Probing Visual Reasoning of LVLMs via Counting." In *European Conference on Computer Vision, 2026.*

[ICML' 26] Zeru Shi, Zhenting Wang, Fan Yang, Qifan Wang, **Ruixiang Tang**†. "A Single Layer to Explain Them All: Understanding Massive Activations in Large Language Models." In *International Conference on Machine Learning, 2026.*

[ICML' 26] Jiaqian Li*, Yanshu Li*, Ligong Han, **Ruixiang Tang**, Wenya Wang. "Train Once, Reuse Everywhere: Generalizable Implicit In-Context Learning by Routing Attention." In *International Conference on Machine Learning, 2026.*

[ACL' 26] Can Jin*, Rui Wu*, Tong Che*, Qixin Zhang, Hongwu Peng, Jiahui Zhao, Zhenting Wang, Wenqi Wei, Ligong Han, Zhao Zhang, Yuan Cao, **Ruixiang Tang**†, Dimitris N Metaxas†. "Reasoning over Precedents Alongside Statutes: Case-Augmented Deliberative Alignment for LLM Safety." In *Annual Meeting of the Association for Computational Linguistics, ACL Main, 2026.*

[ICME' 26] Yanchuan Tang*, Taowen Wang*, Yuefei Chen, Boxuan Zhang, Qiang Guan, **Ruixiang Tang**†. "Shifting Uncertainty to Critical Moments: Towards Reliable Uncertainty Quantification for

VLA Model." In *IEEE International Conference on Multimedia and Expo, 2026.* (**Best Paper Candidate**)

[CVPR' 26] Yuefei Chen, Jiang Liu, Xiaodong Lin, **Ruixiang Tang**†. "Distilling Counterfactual Reasoning from Language to Vision: Causal Graph-Guided Post-Training for Video Understanding." In *Conference on Computer Vision and Pattern Recognition, CVPR Findings, 2026.*

[AAAI' 26] Yanshu Li, Jianjiang Yang, Ziteng Yang, Bozheng Li, Ligong Han, Hongyang He, Zhengtao Yao, Yingjie Victor Chen, Songlin Fei, Dongfang Liu, **Ruixiang Tang**†. "Make LVLMs Focus: Context-Aware Attention Modulation for Better Multimodal In-Context Learning." In *AAAI Conference on Artificial Intelligence, 2026.* (**Oral Presentation**)

[AAAI' 26] Yuefei Chen, Vivek K. Singh, Jing Ma, **Ruixiang Tang**†. "CounterBench: Evaluating and Improving Counterfactual Reasoning in Large Language Models." In *AAAI Conference on Artificial Intelligence, 2026.*

[AAAI' 26] Yanshu Li, Jianjiang Yang, Zhennan Shen, Ligong Han, Haoyan Xu, **Ruixiang Tang**†. "CATP: Contextually Adaptive Token Pruning for Efficient and Enhanced Multimodal In-Context Learning." In *AAAI Conference on Artificial Intelligence, 2026.* (**Oral Presentation**)

[ICLR' 26] Runjia Zeng, Qifan Wang, Qiang Guan, **Ruixiang Tang**, Lifu Huang, Zhenting Wang, Xueling Zhang, Cheng Han, Dongfang Liu. "TokenSeek: Memory Efficient Fine Tuning via Instance-Aware Token Ditching." In *International Conference on Learning Representations, 2026.*

[EACL' 26] Yu-Neng Chuang†, Guanchu Wang, Chia-Yuan Chang, **Ruixiang Tang**, Shaochen Zhong, Fan Yang, Andrew Wen, Mengnan Du, Xuanting Cai, Vladimir Braverman, Xia Hu. "FaithLM: Towards Faithful Explanations for Large Language Models." In *European Chapter of the Association for Computational Linguistics, 2026.*

[TKDD' 25] Yu-Neng Chuang, Kwei-Herng Lai, **Ruixiang Tang**, Mengnan Du, Chia-Yuan Chang, Na Zou, Xia Hu. "Fair-RGNN: Mitigating Relational Bias on Knowledge Graphs." In *Transactions on Knowledge Discovery from Data, 2025.*

[EMNLP' 25] Yanshu Li, Jianjiang Yang, Tian Yun, Pinyuan Feng, Jinfa Huang, **Ruixiang Tang**†. "TACO: Enhancing Multimodal In-context Learning via Task Mapping-Guided Sequence Configuration." In *Empirical Methods in Natural Language Processing, EMNLP Main, 2025.*

[EMNLP' 25] Hongyi Liu, Shaochen Zhong, Xintong Sun, Minghao Tian, Mohsen Hariri, Zirui Liu, **Ruixiang Tang**, Zhimeng Jiang, Jiayi Yuan, Yu-Neng Chuang, Li Li, Soo-Hyun Choi, Rui Chen, Vipin Chaudhary, Xia Hu. "LoRATK: LoRA Once, Backdoor Everywhere in the Share-and-Play Ecosystem." In *Findings of Empirical Methods in Natural Language Processing, 2025.*

[EMNLP' 25] Lingxi Zhang, Yu-Neng Chuang, Guanchu Wang, **Ruixiang Tang**, Xuanting Cai, Rajesh Shenoy, Xia Hu. "A Decoupled Multi-Agent Framework for Complex Text Style Transfer." In *Findings of Empirical Methods in Natural Language Processing, 2025.*

[ICCV' 25] Taowen Wang, Cheng Han, James Liang, Wenhao Yang, Dongfang Liu, Luna Xinyu Zhang, Qifan Wang, Jiebo Luo, **Ruixiang Tang**†. "Exploring the Adversarial Vulnerabilities of Vision-Language-Action Models in Robotics." In *International Conference on Computer Vision, 2025.*

[ICCV' 25] Liwei Che, Tony Qingze Liu, Jing Jia, Weiyi Qin, **Ruixiang Tang**, Vladimir Pavlovic. "Hallucinatory Image Tokens: A Training-free EAZY Approach to Detecting and Mitigating Object Hallucinations in LVLMs." In *International Conference on Computer Vision, 2025.*

[ACL' 25] Huaizhi Ge, Yiming Li, Qifan Wang, Yongfeng Zhang, **Ruixiang Tang**†. "When Backdoors Speak: Understanding LLM Backdoor Attacks Through Model-Generated Explanations." In *Annual*

Meeting of the Association for Computational Linguistics, ACL Main, 2025.

[ACL' 25] Mingyu Jin, Weidi Luo, Sitao Cheng, Xinyi Wang, Wenyue Hua, **Ruixiang Tang**, William Yang Wang, Yongfeng Zhang. "Disentangling Memory and Reasoning Ability in Large Language Models." In *Annual Meeting of the Association for Computational Linguistics, ACL Main, 2025.*

[AACL' 25] Haein Kong, A M Muntasir Rahman, **Ruixiang Tang**, Vivek Singh. "SafePersuasion: A Dataset, Taxonomy, and Baselines for Analysis of Rational Persuasion and Manipulation." In *Findings of IJCNLP-AACL, 2025.*

[BioAdv' 25] Matthew Neeley*, Guantong Qi*, Guanchu Wang*, **Ruixiang Tang**, Dongxue Mao, Chaozhong Liu, Sasidhar Pasupuleti, Bo Yuan, Fan Xia, Pengfei Liu, Zhandong Liu, Xia Hu. "Survey and Improvement Strategies for Gene Prioritization with Large Language Models." In *Bioinformatics Advances, 2025.*

[CIKM' 25] Qingcheng Zeng, Mingyu Jin, Qinkai Yu, Zhenting Wang, Wenyue Hua, Guangyan Sun, Yanda Meng, Shiqing Ma, Qifan Wang, Felix Juefei-Xu, Fan Yang, Kaize Ding, **Ruixiang Tang**, Yongfeng Zhang. "Uncertainty Quantification for Multiple-Choice Questions is Just One-Token Deep." In *ACM Conference on Information and Knowledge Management, 2025.*

[COLM' 25] Yanshu Li*, Yi Cao*, Hongyang He, Qisen Cheng, Xiang Fu, Xi Xiao, Tianyang Wang, **Ruixiang Tang**†. "M^2IV: Towards Efficient and Fine-grained Multimodal In-Context Learning via Representation Engineering." In *Conference on Language Modeling, 2025.*

[ICML' 25] Mingyu Jin*, Kai Mei*, Wujiang Xu, Mingjie Sun, **Ruixiang Tang**, Mengnan Du, Zirui Liu, Yongfeng Zhang. "Massive Values in Self-Attention Modules are the Key to Contextual Knowledge Understanding." In *International Conference on Machine Learning, 2025.*

[ICLR' 25] Yiyang Liu*, James Chenhao Liang*, **Ruixiang Tang**, Yugyung Lee, Majid Rabbani, Sohail Dianat, Raghuveer Rao, Lifu Huang, Dongfang Liu, Qifan Wang, Cheng Han. "Re-Imagining Multimodal Instruction Tuning: A Representation View." In *International Conference on Learning Representations, 2025.*

[KDD Explorations' 25] Zihao Li, **Ruixiang Tang**, Lu Cheng, Shuaiqiang Wang, Dawei Yin, Mengnan Du. "DBR: Divergence-Based Regularization for Debiasing Natural Language Understanding Models." In *ACM SIGKDD Explorations Newsletter, 2025.*

[CACM' 24] **Ruixiang Tang**, Yu-Neng Chuang, Xia Hu. "The Science of Detecting LLM-Generated Text." In *Communications of the ACM, 2024.* (**Featured Cover Article**)

[JBI' 24] Yu-Neng Chuang, **Ruixiang Tang**, Xiaoqian Jiang, Xia Hu. "SPeC: A Soft Prompt-Based Calibration on Performance Variability of Large Language Model in Clinical Notes Summarization." In *Journal of Biomedical Informatics, 2024.*

[EMNLP' 24] Guanchu Wang*, Yu-Neng Chuang*, **Ruixiang Tang**, Shaochen Zhong, Jiayi Yuan, Hongye Jin, Zirui Liu, Vipin Chaudhary, Shuai Xu, James Caverlee, Xia Hu. "Taylor Unswift: Secured Weight Release for Large Language Models via Taylor Expansion." In *Empirical Methods in Natural Language Processing, EMNLP Main, 2024.*

[EMNLP' 24] Yuqing Zhou, **Ruixiang Tang**, Ziyu Yao, Ziwei Zhu. "Navigating the Shortcut Maze: A Comprehensive Analysis of Shortcut Learning in Text Classification by Language Models." In *Findings of Empirical Methods in Natural Language Processing, 2024.*

[EMNLP' 24] Wenyue Hua, Xianjun Yang, Mingyu Jin, Zelong Li, Wei Cheng, **Ruixiang Tang**, Yongfeng Zhang. "TrustAgent: Towards Safe and Trustworthy LLM-based Agents." In *Findings of Empirical Methods in Natural Language Processing, 2024.*

[NAACL' 24] **Ruixiang Tang***, Yu-Neng Chuang*, Xuanting Cai, Mengnan Du, Xia Hu. "Secure Your Model: An Effective Key Prompt Protection Mechanism for Large Language Models." In *Findings of Annual Conference of the North American Chapter of the Association for Computational Linguistics, 2024.*

[TKDD' 24] Jingfeng Yang*, Hongye Jin*, **Ruixiang Tang***, Xiaotian Han*, Qizhang Feng*, Haoming Jiang, Shaochen Zhong, Bing Yin, Xia Hu. "Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond." In *Transactions on Knowledge Discovery from Data, 2024.*

[EMNLP' 23] **Ruixiang Tang**, Gord Lueck, Rodolfo Quispe, Huseyin A Inan, Janardhan Kulkarni, Xia Hu. "Assessing Privacy Risks in Language Models: A Case Study on Summarization Tasks." In *Findings of Empirical Methods in Natural Language Processing, 2023.*

[NeurIPS' 23] **Ruixiang Tang***, Jiayi Yuan*, Yiming Li, Zirui Liu, Rui Chen, Xia Hu. "Setting the Trap: Capturing and Defeating Backdoors in Pretrained Language Models through Honeypots." In *Conference on Neural Information Processing Systems, 2023.*

[NeurIPS' 23] Zirui Liu*, Guanchu Wang*, Shaochen Zhong, Zhaozhuo Xu, Daochen Zha, **Ruixiang Tang**, Zhimeng Jiang, Kaixiong Zhou, Vipin Chaudhary, Shuai Xu, Xia Hu. "Winner-Take-All Column Row Sampling for Memory Efficient Adaptation of Language Model." In *Conference on Neural Information Processing Systems, 2023.*

[AMIA' 23] **Ruixiang Tang***, Xiaotian Han*, Xiaoqian Jiang, Xia Hu. "Does Synthetic Data Generation of LLMs Help Clinical Text Mining?" In *AMIA Annual Symposium, 2023.* (**Poster**)

[AMIA' 23] Jiayi Yuan, **Ruixiang Tang**, Xiaoqian Jiang, Xia Hu. "Large Language Models for Healthcare Data Augmentation: An Example on Patient-Trial Matching." In *AMIA Annual Symposium, 2023.* (**Best Student Paper Award**)

[AMIA' 23] Syed A. Rizvi, **Ruixiang Tang**, Xiaoqian Jiang, Xiaotian Ma, Xia Hu. "Local Contrastive Learning for Medical Image Recognition." In *AMIA Annual Symposium, 2023.*

[ACL' 23] **Ruixiang Tang**, Dehan Kong, Longtao Huang, Hui Xue. "Large Language Models Can be Lazy Learners: Analyze Shortcuts in In-Context Learning." In *Findings of Annual Meeting of the Association for Computational Linguistics, 2023.*

[CIKM' 23] **Ruixiang Tang***, Hongye Jin*, Mengnan Du, Curtis Wigington, Rajiv Jain, Xia Hu. "Exposing Model Theft: A Robust and Transferable Watermark for Thwarting Model Extraction Attacks." In *ACM Conference on Information and Knowledge Management, short paper, 2023.*

[CIKM' 23] Yu-Neng Chuang, Guanchu Wang, Chia-Yuan Chang, Kwei-Herng Lai, Daochen Zha, **Ruixiang Tang**, Fan Yang, Alfredo Costilla Reyes, Kaixiong Zhou, Xiaoqian Jiang, Xia Hu. "DiscoverPath: A Knowledge Refinement and Retrieval System for Interdisciplinarity on Biomedical Research." In *ACM Conference on Information and Knowledge Management, demo paper, 2023.* (**Best Demo Paper Honorable Mention**)

[ICHI' 23] Shenghan Zhang*, Haoxuan Li*, **Ruixiang Tang***, Sirui Ding*, Laila Rasmy, Degui Zhi, Na Zou, Xia Hu. "PheME: A Deep Ensemble Framework for Improving Phenotype Prediction from Multi-modal Data." In *IEEE International Conference on Healthcare Informatics, 2023.*

[ECML' 23] **Ruixiang Tang**, Mengnan Du, Xia Hu. "Deep Serial Number: Computational Watermark for DNN Intellectual Property Protection." In *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, 2023.*

[KDD Explorations' 23] **Ruixiang Tang**, Qizhang Feng, Ninghao Liu, Fan Yang, Xia Hu. "Did You Train on My Dataset? Towards Public Dataset Protection with Clean-Label Backdoor Watermarking."

In *ACM SIGKDD Explorations Newsletter, 2023.*

[TCSS' 22] Fangsheng Wu, Mengnan Du, Chao Fan, **Ruixiang Tang**, Yang Yang, Ali Mostafavi, Xia Hu. "Understanding Social Biases Behind Location Names in Contextual Word Embedding Models." In *IEEE Transactions on Computational Social Systems, 2022.*

[ICLR' 22] Qizhang Feng, Ninghao Liu, Fan Yang, **Ruixiang Tang**, Mengnan Du, Xia Hu. "DEGREE: Decomposition Based Explanation for Graph Neural Networks." In *International Conference on Learning Representations, 2022.*

[FBD' 22] **Ruixiang Tang***, Ninghao Liu*, Fan Yang, Na Zou, Xia Hu. "Defense Against Explanation Manipulation." In *Frontiers in Big Data, 2022.*

[AMIA' 22] Sirui Ding, **Ruixiang Tang**, Daochen Zha, Na Zou, Kai Zhang, Xiaoqian Jiang, Xia Hu. "Fairly Predicting Graft Failure in Liver Transplant for Organ Assigning." In *AMIA Annual Symposium, 2022.* (**Best Student Paper Finalist**)

[AAAI' 22] Mengnan Du, **Ruixiang Tang**, Weijie Fu, Xia Hu. "Towards Debiasing DNN Models From Spurious Feature Influence." In *AAAI Conference on Artificial Intelligence, 2022.*

[WWW' 21] **Ruixiang Tang**, Mengnan Du, Yuening Li, Zirui Liu, Na Zou, Xia Hu. "Mitigating Gender Bias in Captioning Systems." In *The Web Conference, 2021.*

[NeurIPS' 21] Mengnan Du, Subhabrata Mukherjee, Guanchu Wang, **Ruixiang Tang**, Ahmed Awadallah, Xia Hu. "Fairness via Representation Neutralization." In *Conference on Neural Information Processing Systems, 2021.*

[KDD' 20] **Ruixiang Tang**, Mengnan Du, Ninghao Liu, Fan Yang, Xia Hu. "An Embarrassingly Simple Approach for Trojan Attack in Deep Neural Networks." In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2020.*

[ISBI' 20] Chunze Lin, **Ruixiang Tang**, Darryl D Lin, Langechuan Liu, Jiwen Lu, Yunqiang Chen, Dashan Gao, Jie Zhou. "Deep Feature Disentanglement Learning for Bone Suppression in Chest Radiographs." In *IEEE International Symposium on Biomedical Imaging, 2020.*

[SPIE' 19] **Ruixiang Tang**, Fakrul Islam Tushar, Songyue Han, Rui Hou, Geoffrey D Rubin, Joseph Y Lo. "Classification of Chest CT Using Case-level Weak Supervision." In *SPIE Medical Imaging: Computer-Aided Diagnosis, 2019.*

[SPIE' 19] Wanyi Fu, Shobhit Sharma, Taylor Smith, Rui Hou, Ehsan Abadi, Vignesh Selvakumaran, **Ruixiang Tang**, Joseph Y Lo, W Paul Segars, Anuj J Kapadia, Justin B Solomon, Geoffrey D Rubin, Ehsan Samei. "Multi-organ Segmentation in Clinical-Computed Tomography for Patient-specific Image Quality and Dose Metrology." In *SPIE Medical Imaging: Physics of Medical Imaging, 2019.*

Preprint and Under Review

[Preprint' 26] Boxuan Zhang, Jianing Zhu, Xiaotian Han, **Ruixiang Tang**†. "Dream2Discern: Learning from Hard Latent Variants for Runtime Failure Prediction in Multi-Agent Systems."

[Preprint' 26] Guangxun Zhang*, Brian Cai*, Boxuan Zhang, Chao Chen†, **Ruixiang Tang**†. "JIVE: Jacobian-Informed Volume Expansion for Diverse Generative Sampling."

[Preprint' 26] Minghao Guo, Qingyue Jiao, Zeru Shi, Yihao Quan, Boxuan Zhang, Danrui Li, Liwei Che, Wujiang Xu, Shilong Liu, Zirui Liu, Mubbasir Kapadia, Vladimir Pavlovic, Jiang Liu, Mengdi

Wang, Yiyu Shi, Dimitris N. Metaxas, **Ruixiang Tang**†. "MemEye: A Visual-Centric Evaluation Framework for Multimodal Agent Memory."

[Preprint' 26] Boxuan Zhang, Jianing Zhu, Qifan Wang, Jiang Liu, **Ruixiang Tang**†. "Micro-Defects Expose Macro-Fakes: Detecting AI-Generated Images via Local Distributional Shifts."

[Preprint' 26] Zeru Shi, Kai Mei, Yihao Quan, Dimitris N. Metaxas, **Ruixiang Tang**†. "Improving Visual Reasoning with Iterative Evidence Refinement."

[Preprint' 26] Yuefei Chen, Yihao Quan, Xiaodong Lin, **Ruixiang Tang**†. "Where Fake Citations Are Made: Tracing Field-Level Hallucination to Specific Neurons in LLMs."

[Preprint' 26] Yanshu Li, **Ruixiang Tang**, Junyi Jessy Li, Elias Stengel-Eskin. "ActICL: Post-Training for Agentic Multimodal In-Context Learning."

[Preprint' 26] Zeru Shi*, Jindong Jiang*, Ryo Hachiuma, Yonggan Fu, Byung-Kwan Lee, Peter Belcak, Saurav Muralidharan, Ayush Agarwal, **Ruixiang Tang**, Ambrish Dantrey, Pavlo Molchanov, Anjul Patney, Subhashree Radhakrishnan. "EdgeWise: Bringing Long-Horizon Agents to the Edge via Selective Cloud Delegation."

[Preprint' 26] Liwei Che, Yihao Quan, Sen Fang, Hongyi Wang, Ranjay Krishna, **Ruixiang Tang**†, Vladimir Pavlovic†. "Why MLLMs Struggle to Count: Overcoming Individuation and Aggregation Bottlenecks with ConvStack."

[Preprint' 26] Runjia Zeng, Priyabrata Senapati, **Ruixiang Tang**, Dongfang Liu, Qiang Guan. "Q-Bridge: Code Translation for Quantum Machine Learning via LLMs."

[Preprint' 26] Shaochen Zhong, Jiamu Zhang, Hoang Anh Duy Le, Wenya Xie, Yifan Lu, Xintong Sun, Mohsen Hariri, Hongyi Liu, Guanchu Wang, Zhaozhuo Xu, Zirui Liu, Shuai Xu, Ning Xie, Li Li, Rui Chen, **Ruixiang Tang**, Xia Hu, Vipin Chaudhary. "Sweeping Promptable Spoofs under the DirtyRAG: A Practical, Query-Blind RAG Attack Done Right."

[Preprint' 26] Xiang Zhang, Boxuan Zhang, Alireza Naghizadeh, Mohab Mohamed, Dongfang Liu, **Ruixiang Tang**, Dimitris Metaxas. "Data Augmentation for High-Fidelity Generation of CAR-T/NK Immunological Synapse Images."

[Preprint' 25] Zeru Shi, Yingjia Wan, Zhenting Wang, Qifan Wang, Fan Yang, Elisa Kreiss, **Ruixiang Tang**†. "Meaningless Tokens, Meaningful Gains: How Activation Shifts Enhance LLM Reasoning."

[Preprint' 25] Zicong He, Boxuan Zhang, Weihao Liu, **Ruixiang Tang**, Lu Cheng. "What Shapes a Creative Machine Mind? Comprehensively Benchmarking Creativity in Foundation Models."

[Preprint' 25] Zeru Shi, Zhenting Wang, Yongye Su, Weidi Luo, Fan Yang, **Ruixiang Tang**, Yongfeng Zhang. "Auto-Prompt Generation is Not Robust: Prompt Optimization Driven by Pseudo Gradient."

RESEARCH FUNDING

Amazon Research Award *2026*
Sponsor: Amazon, Amazon Research Awards (Spring 2026 call)
Role: Principal Investigator
Status: Awarded, October 2026

T1DKGent: A Multi-Agent Framework for Constructing Temporally Evolving T1D Knowledge Graphs *2026*
Sponsor: National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), National Institutes of Health, through the PanKbase Opportunity Fund

Role: Contact Principal Investigator
Multiple Principal Investigator: Zhiping Pang, Robert Wood Johnson Medical School, Rutgers University
Consultant: Yan Li, Case Western Reserve University
Prime Award: University of Michigan (Steve Parker, Principal Investigator); Rutgers subaward
Amount: $142,696
Project Period: January 1, 2026 – December 31, 2026
Status: Awarded, March 2026

AI–Geoscience Collaborations for a Changing Arctic *2026*
Sponsor: Rutgers School of Arts and Sciences, Research in Academic Themes (RIAT)
Role: Co-Principal Investigator
Principal Investigator: Åsa Rennermalm, Rutgers University
Co-Principal Investigator: Yongfeng Zhang, Department of Computer Science, Rutgers University
Amount: $70,000
Status: Awarded, January 15, 2026

AMD University Program: AI & HPC Cluster Allocation *2026*
Sponsor: AMD University Program (in-kind computing award)
Role: Principal Investigator
Amount: 1,500 GPU hours on the AMD AI & HPC Cluster
Status: Awarded, 2026

Rutgers Climate and Energy Fellowships to Support CS–Geography Collaboration *2025*
Sponsor: Rutgers Climate and Energy Institute (RCEI) Fellowship Program
Role: Co-Recipient (Computer Science)
Amount: $15,000 ($7,500 Computer Science; $7,500 Geography)
Status: Awarded, 2025

OpenAI Researcher Access Program *2025*
Sponsor: OpenAI (in-kind API credits)
Role: Recipient
Amount: $2,000 in API credits
Status: Awarded, 2025

TEACHING EXPERIENCE

Instructor, Department of Computer Science, Rutgers University *Spring 2026*
CS 439 Introduction to Data Science

Instructor, Department of Computer Science, Rutgers University *Fall 2025*
CS 550 Massive Data Mining

Instructor, Department of Computer Science, Rutgers University *Spring 2025*
CS 439 Introduction to Data Science

Instructor, Department of Computer Science, Rutgers University *Fall 2024*
CS 598 Trustworthy AI Foundations

INVITED TALKS

SRI International *November 2026*
Invited research talk introducing the group's recent work on trustworthy and reliable AI; hosted by Dr. Rakesh Kumar.

Auburn University, COMP 5630/6630 Machine Learning *October 2026*
Invited guest lecture: "When AI Fails: From Mechanistic Understanding to Proactive Auditing."

Rutgers Young Scholars Program in Discrete Mathematics (DIMACS) *July 2026*
Invited outreach lecture introducing artificial intelligence and computer vision to selected high school students.

IBM *June 2026*
Invited research talk on agent safety and early failure detection: "Before the Failure: Auditing and Training Reliable Agentic Systems."

SIAM International Conference on Data Mining (SDM) *May 2025*
Minitutorial: "Exploitation and Mitigation: Understanding Large-Scale Machine Learning Robustness under Paradigm Shift," co-presented with Xi Li (University of Alabama at Birmingham) and Muchao Ye (University of Iowa).

SAPA Science & Careers Symposium, Sino-American Pharmaceutical Professionals Association *April 2025*
Opening technical talk of the Agentic AI Workshop: "Opening the Black Box of Large Language Models."

New Jersey Institute of Technology, Department of Data Science *October 2024*
Department seminar: "Safe and Reliable Large Language Models."

MENTORSHIP

Ph.D. Students

Qiyuan Shi, Rutgers University, *September 2026 - Present*
Project: Model Steering

Boxuan Zhang, Rutgers University, *September 2025 - Present*
Project: Agent Safety, Agent Memory

Zeru Shi, Rutgers University, *September 2025 - Present*
Project: Mechanistic Interpretation; Reinforcement Learning; LLM/VLM Reasoning

Yihao Quan, Rutgers University, *September 2025 - Present*
Project: Hallucination Detection/Mitigation; Spatial Reasoning; Unified Model

Yuefei Chen (Co-advised), Rutgers University, *September 2024 - Present*
Project: Large language model for causal inference

Master Students

Yuqi Pu, Rutgers University, *January 2026 - Present*
Project: AI for Geoscience

Yanshu Li, Brown University, *January 2025 - May 2026*
Project: Vision In-context Learning (AAAI 2026, COLM 2025, EMNLP 2025)

Huaizhi Ge, Columbia University, *September 2024 - September 2025*
Project: Defending Large Language Models Backdoor Attacks (ACL 2025)

Undergraduate Students

Bobby (Yanchuan) Tang, University of Wisconsin–Madison, *September 2025 - Present*
Project: Uncertainty Quantification for Vision-Language-Action Models (ICME 2026)

Rui Wu, Rutgers University, *September 2024 - Present*
Project: LLM Safety Alignment

Leroy Souz, Rutgers University, *September 2024 - May 2026*
Project: Detecting AI-generated Audio (Semifinal for the 2025 Rice Business Plan Competition.)

Syed A. Rizvi, University of Houston *August 2022 - May 2023*
Current: Enrolled in the Ph.D. program in Computer Science at Yale University
Project: Medical Image-text Pretraining, Interpretation, AMIA 2023

Ph.D. Dissertation Committees

Zihao Xu, Ph.D. in Computer Science, Rutgers University *2026*
Wenjie Qiu, Ph.D. in Computer Science, Rutgers University *2026*
Zhizhi Wang, Ph.D. in Computer Science, Rutgers University *2026*
Minghao Guo, Ph.D. in Computer Science, Rutgers University *2026*
Jindong Jiang, Ph.D. in Computer Science, Rutgers University *2025*
Zhenting Wang, Ph.D. in Computer Science, Rutgers University *2025*
Shiyu Zhao, Ph.D. in Computer Science, Rutgers University *2025*
Chang Chen, Ph.D. in Computer Science, Rutgers University *2025*
Kunpeng Song, Ph.D. in Computer Science, Rutgers University *2025*

MEDIA COVERAGE AND INTERVIEWS

"Rutgers Undergraduate Earns First-Author Acceptance at COLM 2026" Rutgers Computer Science News *2026*

"Rutgers Computer Science Research Team Awarded AMD AI & HPC Cluster Compute Resources" Rutgers Computer Science News *2026*

"Rutgers Climate and Energy Fellowships provide support for early career researchers to participate in RCEI program offerings" Rutgers Climate and Energy Institute *2025*

"AI vs. copyright: How large vision-language models are changing IP protection" Devdiscourse *2025*

"2025 SAPA Science and Careers Symposium" Sino-American Pharmaceutical Professionals Association *2025*

"The Science of Detecting LLM-Generated Text" Communications of the ACM *2024*

"Review of 'The Science of Detecting LLM-Generated Text'" by Gunnar Wolf, Computing Reviews CR147857 *2024*

"New Faculty Spotlight: Ruixiang (Ryan) Tang (Computer Science)" Rutgers Division of Mathematical and Physical Sciences *2024*

"Ruixiang Tang: Detecting LLM-Generated Texts" Research interview, NiklasOPF (YouTube) *2024*

"Rice CS' Xia Ben Hu investigates LLMs and likely applications" Rice CS News *2023*

"Using ChatGPT to Generate Data for Medical AI Models" Tianqiao and Chrissy Chen Institute (TCCI) Meeting Report *2023*

"Mitigating Gender Bias in Captioning Systems" NSF Fairness in AI Program *2021*

"TrojanNet – a simple yet effective attack on machine learning models", The Daily Swig *2020*

SELECTED ACADEMIC SERVICES

Associate Editor, Smart Health Journal

Guest Editor, Pattern Recognition, Special Issue on "World Modeling for Dynamic Video Understanding"

Tutorial Co-Chair, ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB 2027), University of Massachusetts Boston, August 10–13, 2027

Area Chair: ACL Rolling Review (ARR), Association for Computational Linguistics, 2025–2026 (October 2025, January 2026, May 2026 cycles)

Area Chair, International Conference on Learning Representations (ICLR 2027)

Senior Program Committee Member, ACM International Conference on Information and Knowledge Management (CIKM 2023, 2024, 2025, 2026)

Senior Program Committee Member, AAAI Conference on Artificial Intelligence (AAAI 2027)

Special Session Organizer, IEEE International Conference on Multimedia and Expo (ICME 2026), Special Session on "Trustworthy and Secure Vision-Language-Action Models: Foundations, Methods, and Evaluation"

Research Mentor: ML Alignment & Theory Scholars (MATS), since September 2026; mentoring fellows on AI control and monitoring, and interpretability

Reviewing:

ICLR (Area Chair); AAAI (Senior PC)	*2027*
ACL ARR (Area Chair); CIKM (Senior PC); ICLR; ICML; CVPR; BMVC; COLM; KDD AI4Science; ECCV; IEEE IV	*2026*
ACL ARR (Area Chair); NeurIPS; CIKM (Senior PC); AAAI (PC); ICML; ICCV; ICLR	*2025*
ICML; ACL ARR; ICLR; CIKM (Senior PC)	*2024*
NeurIPS; CIKM (Senior PC)	*2023*

Journal Reviewer: Proceedings of the National Academy of Sciences (PNAS); Nature Human Behaviour; npj Digital Medicine; Patterns (Cell Press); Cell Reports Methods (Cell Press); Journal of the American Medical Informatics Association (JAMIA); ACM Computing Surveys; IEEE Transactions on Information Forensics and Security; IEEE Transactions on Multimedia; ACM Transactions on Intelligent Systems and Technology; Data Mining and Knowledge Discovery; Decision Support Systems

Recognition: Top Reviewer of NeurIPS 2023

INDUSTRIAL RESEARCH EXPERIENCE

Applied Scientist Intern at Microsoft, USA *May 2023-August 2023*
Project: Medical Dialogue Generation, Personal Identifying Information (PII) Detection.
Mentors: Gord Lueck, Rodolfo Quispe, Huseyin Inan, and Janardhan Kulkarni.

Research Intern at Microsoft Research, USA. *May 2022-August 2022*
Project: Medical Dialogue Summarization, Privacy Risk Analysis in Large Language Models.
Mentors: Gord Lueck, Rodolfo Quispe, Huseyin Inan, and Janardhan Kulkarni.

Research Scientist Intern at Adobe Research, USA *May 2021-August 2021*
Project: Watermarking Computer Vision Foundation Models and APIs.
Mentors: Curtis Wigington and Rajiv Jain.